

 OR-AS Operations Research Applications and Solutions	Case Name: Drainage Culverts Construction Project Owner: Eder Omar Vilca Carrillo	Sector	Construction	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
Processed by	Zahra Hussaini		One of nine std. scenarios	
Date	2025	Project Control	User defined distributions	
File Name	C2025-20		Automatic tracking	
			Tracking based on user input	

1. Project description

Project authenticity

This project describes the different activities and their relations in the construction and improvement of drainage culverts (including corrugated metal pipes and concrete headwalls) along the Puerto La Balsa - Rafael Belaunde route, incorporating COVID-19 safety measures to ensure worker protection and infrastructure functionality. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	34	Serial/Parallel (SP)	67%
Planned Duration (PD)	45 days	Activity Distribution (AD)	60%
Budget At Completion (BAC)	302383.41 EUR	Length of Arcs (LA)	0%
Renewable Resources	6 Types	Topological Float (TF)	50%
Consumable Resources	0 Types		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	12.6	14.7	4.3
CRI-rho	14.0	15.9	3.6
CRI-tau	13.7	19.4	3.5

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	23.3	11.3	5.2
CRI-rho	23.3	11.3	5.2
CRI-tau	23.3	11.3	5.2

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	42.3	31.2	-0.2
SI	47.6	31.1	-0.4
SSI	3.6	3.9	1.6
CRI-r	11.2	13.8	4.3
CRI-rho	13.5	14.3	3.1
CRI-tau	12.8	19.1	3.4

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	20.3	19.9	1	0.4	0.0
PV - SPI	29.2	28.8	CPI	0.4	0.1
PV - SCI	29.3	29.0	SPI	9.6	9.5
ED - 1	17.2	16.7	SPI(t)	11.4	11.4
ED - SPI	28.0	27.6	SCI	9.6	9.5
ED - SCI	28.1	27.7	SCI(t)	11.5	11.4
ES - 1	20.3	20.1	0.8 CPI + 0.2 SPI	3.1	3.1
ES - SPI(t)	31.3	31.2	0.8 CPI + 0.2 SPI(t)	4.7	4.7
ES - SCI(t)	31.4	31.2			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 2 tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0	-44272.8	-84	1	0.7	0.7	1
std dev	0	62611.2	39.6	0	0.4	0.1	0
final	0	0	-112	1	1.0	0.8	1

2.3.3.2. Time forecasting

PD	45 days
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Real Duration	59 days
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Late	31%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	51	9.9	6.8	6.8
PV - SPI	71	38.2	10.2	-10.2
PV - SCI	71	38.2	10.2	-10.2
ED - 1	56	2.8	2.5	2.5
ED - SPI	78	28.3	16.1	-16.1
ED - SCI	78	28.3	16.1	-16.1
ES - 1	54.5	4.9	3.8	3.8
ES - SPI(t)	65.5	10.6	5.5	-5.5
ES - SCI(t)	65.5	10.6	5.5	-5.5

2.3.3.3. Cost forecasting

BAC	302383.41€
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Real Cost	302383.41€
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On budget	0%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	302383.4	0	9E-06	-9E-06
CPI	302383.4	0	9E-06	-9E-06
SPI	437575.5	191190.4	44.7	-44.7
SPI(t)	374861.3	102499.1	24.0	-24.0
SCI	437575.5	191190.4	44.7	-44.7
SCI(t)	374861.3	102499.1	24.0	-24.0
0.8 CPI + 0.2 SPI	316252.1	19613.3	4.6	-4.6
0.8 CPI + 0.2 SPI(t)	311988.9	13584.2	3.2	-3.2